



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

15-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,C	B,C	A,B,C	A,B,D	B,C,D	A,D	A,C
SECTION-II (i)	Q.	1	2	3	4				
	A.	20.00	1.98 to 2.00	9.00	5.17 to 5.20				
SECTION-II (ii)	Q.	5	6	7	8				
	A.	22.56	16.00	0.40	3.00				
SECTION-IV	Q.	1	2						
	A.	A->R,B->P,C->Q,D->R	A->RS,B->RS,C->PQT,D->PS						

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,B	B,C,D	A,B,C,D	A,B,C	A,B,C,D	B,C	B,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4				
	A.	0.04	3.00	1.25	4.00				
SECTION-II (ii)	Q.	5	6	7	8				
	A.	0.05	8.00	5.00	5.00				
SECTION-IV	Q.	1	2						
	A.	A->PQT,B->RT,C->RT,D->QST	A->PR,B->QS,C->P,D->PT						

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,C,D	C	A,B,D	B,C,D	A,B	A,D	B,C
SECTION-II (i)	Q.	1	2	3	4				
	A.	3.00	0.25	0.80	0.33				
SECTION-II (ii)	Q.	5	6	7	8				
	A.	0.25	6.00	2.00	1.00				
SECTION-IV	Q.	1	2						
	A.	A->P,B->Q,C->S,D->Q	A->Q,B->R,C->Q,D->P						

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

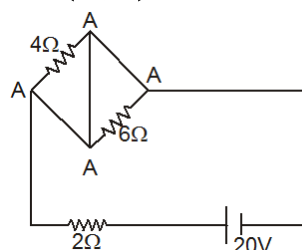
1. Ans (A,C)

$$\frac{GMm}{2r^2} \Delta r = CP v^2;$$

$$2\pi r \left(\text{where, } v = \sqrt{\frac{GM}{r}} \right)$$

$$\rho = \frac{m \Delta r}{4\pi r^2 C}$$

2. Ans (A,C)



there is zero potential difference across 4Ω and 6Ω resistance.

$$i = \frac{20}{2} = 10A$$

power by battery

$$p_b = \varepsilon i = 20 \times 10 = 200 W$$

3. Ans (B,C)

When switch with position (1) is closed.

$$\text{Charge flow} = \frac{2CE}{3}$$

Now switch shifted to position (2)

Charge flow

$$4E \cdot \frac{C}{2} = 2CE$$

$$\text{Heat} = \frac{1}{2} \times 2CE \times 4E = 4CE^2$$

After switch shifted to position (2)

Total heat from starting

$$= \frac{CE^2}{3} + 4CE^2 = \frac{13CE^2}{3}$$

6. Ans (B,C,D)

(A) Electric potential due to dipole = $\frac{k\vec{P} \cdot \vec{r}}{r^3}$

(B) Electric dipole moment is a vector quantity.

(C) Electric potential at equatorial plane is zero.

(D) $\vec{\tau} = \vec{P} \times \vec{E}$

7. Ans (A,D)

(A) If battery is connected, then $E = \frac{V}{d}$
(Constant)

(B) If battery is disconnected, then Q on capacitor will remain same $\Rightarrow$ Electric field decreases.

(C) Since initially capacitor was evacuated (filled with air), its dielectric strength on insertion of dielectric decreases.

(D) $C = \frac{k\epsilon_0 A}{d}$ which increases on insertion of dielectric.

8. Ans (A,C)

Using lens formula and data given in graph.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \dots(1)$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{-10}$$

$$f = +5$$

By differentiating eq. (1)

$$\Rightarrow \frac{-\Delta f}{f^2} = \frac{-\Delta v}{v^2} + \frac{\Delta u}{u^2}$$

$$= \frac{1}{+v^2} \Delta v + \frac{1}{u^2} \Delta v + \frac{\Delta f}{5^2}$$

$$\Rightarrow \frac{1 \times (0.1)}{+10^2} + \frac{1 \times (0.1)}{10^2}$$

$$\Delta f = \frac{0.2}{100} \times 25 = \frac{0.2}{4} = 0.05$$

$$\text{so, } f = 5 \pm 0.05$$

Fractional error is 0.01

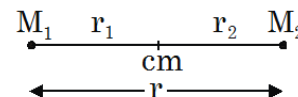
SECTION-II (i)

1. Ans (20.00)

$$r_1 = \frac{M_2 r}{M_1 + M_2}; r_2 = \frac{M_1 r}{M_1 + M_2}$$

$$\frac{GM_1 M_2}{r^2} = \omega^2 r_2$$

$$\omega = \sqrt{\frac{GM_1}{r^2 r_2}}$$



$$T = \frac{2\pi\sqrt{r^2 r_2}}{\sqrt{GM_1}} = \frac{2\pi\sqrt{r^3}}{\sqrt{G(M_1 + M_2)}}$$

$$r = \sqrt[3]{G(M_1 + M_2) \left(\frac{1}{2\pi}\right)^2}$$

2. Ans (1.98 to 2.00)

Actual current is $i = \frac{E}{R}$

Measured current is

$$i = \frac{E}{\frac{R}{10} + \frac{(10R)(R)}{11R}} = 0.99 \frac{E}{R}$$

$$\therefore \% \text{ error} = -1.0\%$$

Actual voltage across R is E

and measured voltage is

$$(i) \left(\frac{10R \times R}{11R} \right) = 0.9 E$$

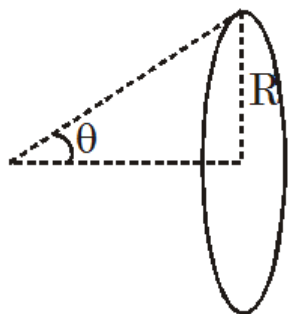
$$\therefore \% \text{ error is } -10\%$$

3. Ans (9.00)

$$F = \frac{\mu_0}{2\pi} \frac{i_1 i_2}{r} = 2 \times 10^{-7} \times \frac{186 \times 24}{1 \times 10^{-4}}$$

$$= 8.928 \approx 9 \text{ N/m}$$

4. Ans (5.17 to 5.20)



$$\frac{\phi}{\phi_T} = \frac{1}{4} = \frac{2\pi(1 - \cos \theta)}{4\pi}$$

$$\cos \theta = \frac{1}{2} \Rightarrow \frac{R}{x_1} = \sqrt{3}$$

$$x_1 = \frac{R}{\sqrt{3}} = 10\sqrt{3} \text{ cm} = 17.32 \text{ cm}$$

$$\frac{\phi}{\phi_T} = \frac{1}{5} = \frac{2\pi}{4\pi}(1 - \cos \theta')$$

$$\cos \theta' = \frac{3}{5} \Rightarrow \theta' = 53^\circ$$

$$\frac{R}{x_2} = \tan 53^\circ = \frac{4}{3}$$

$$x_2 = \frac{3}{4} \times R = 22.5 \text{ cm}$$

$$\Delta x = 5.18 \text{ cm}$$

SECTION-II (ii)

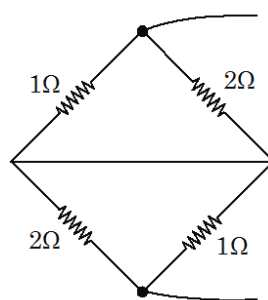
5. Ans (22.56)

Total rotation $\rightarrow$ 22 main scale rotation + 56 circular scale rotation

Length of body = 22.56 mm

6. Ans (16.00)

For time constant



$$R_{eq} = \frac{4}{3}; \tau = \frac{4}{3} \times 4 = \frac{16}{3} \mu\text{s}$$

Charge on capacitor in steady state

$$= 4 \Delta V_{BC} = 4 \times 4 = 16 \mu\text{C}$$

$$\text{Charge as function of time} = 16 \left(1 - e^{-\frac{t}{16/3}} \right)$$

$$I_{BC} = \frac{dQ}{dt} = 16 \left(0 + e^{-\frac{t}{16/3}} \times \frac{3}{16} \right)$$

$$= 3 \left(e^{-\frac{t}{16/3} \mu\text{s}} \right); I_0 \tau = \frac{3 \times 16}{3} = 16$$

7. Ans (0.40)

$$mg = F_{slab} = \frac{dU}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}$$

$$= \frac{1}{2} V^2 \frac{d}{dx} \left(\frac{\epsilon_0}{d} (L^2 + (k-1)Lx) \right)$$

$$= \frac{1}{2} \frac{V^2 (k-1)L}{d}$$

after the slab falls off.

$$2mg - mg = 2ma$$

$$a = \frac{g}{2}; \frac{L}{2} = \frac{1}{2} \times \frac{g}{2} t^2$$

$$t = \sqrt{\frac{2L}{g}} = \sqrt{\frac{2 \times 0.8}{10}} = \sqrt{0.16} = 0.4 \text{ sec}$$

8. Ans (3.00)

$$u = 5.0 \text{ cm}$$

$$v = 10.0 \text{ cm}$$

$$f = \frac{uv}{u+v} = \frac{10}{3} \text{ cm}$$

$$\frac{df}{f} = f \left(\frac{du}{u^2} + \frac{dv}{v^2} \right)$$

$$\frac{df}{f} = f (du) \left(\frac{1}{u^2} + \frac{1}{v^2} \right)$$

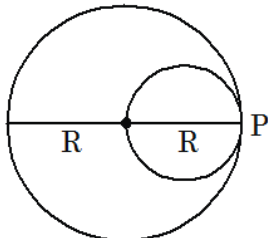
% error in focal length

$$= f (du) \left(\frac{1}{u^2} + \frac{1}{v^2} \right) 100 = \frac{10}{3}$$

SECTION-IV

1. Ans (A->R,B->P,C->Q,D->R)

$$(A) V_{P(R)} = \frac{KQ}{R} = \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi R^3 \rho}{R} = \frac{\rho}{3\epsilon_0} R^2$$

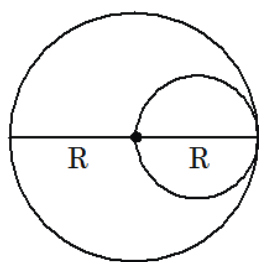


$$V_{P(R/2)} = \frac{\frac{KQ}{8}}{R/2} = \frac{KQ}{4R} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\frac{4}{3}\pi R^3}{4R} = \frac{\rho}{12\epsilon_0} R^2$$

$$\text{So net } V_P = \frac{\rho}{3\epsilon_0} R^2 - \frac{\rho}{12\epsilon_0} R^2 = \frac{3}{4} \cdot \frac{\rho}{3\epsilon_0} R^2 = \frac{\rho R^2}{4\epsilon_0}$$

$$(B) V_{P(R)} \frac{KQ}{R} = \frac{K\sigma 4\pi R^2}{R} = \frac{1}{4\pi\epsilon_0} \cdot \sigma 4\pi R = \frac{\sigma R}{2\epsilon_0}$$

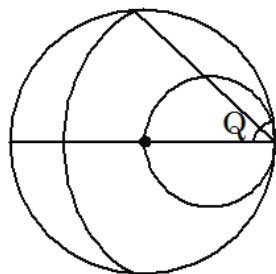
$$V_{P(R/2)} = \frac{\frac{KQ}{4}}{R/2} = \frac{KQ}{2R} = \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2} \cdot \frac{\sigma 4\pi R^2}{R} = \frac{\sigma R}{2\epsilon_0}$$



$$\text{So net } V_P = \frac{\sigma R}{\epsilon_0} + \frac{\sigma R}{2\epsilon_0} = \frac{3\sigma R}{2\epsilon_0}$$

$$(C) V_{P(R)} = 4K\sigma R$$

$$V_{P(R/2)} = 4K\sigma \frac{R}{2}$$



$$\text{So net } V_P = 4K\sigma R - \frac{4K\sigma R}{2} = 2K\sigma R = 2\sigma R \frac{1}{4\pi\epsilon_0} = \frac{\sigma R}{2\pi\epsilon_0}$$

2. Ans (A->RS,B->RS,C->PQT,D->PS)

For D :

$$V_{eq} = \frac{\frac{1.5}{2} + \frac{V}{1}}{\frac{1}{2} + \frac{1}{1}} = \frac{(V + \frac{3}{4})}{\frac{3}{2}} = \frac{2V}{3} + \frac{1}{2}$$

$$i = \frac{\frac{2V}{3} + \frac{1}{2}}{\frac{2}{3} + 4}$$

Similar for C

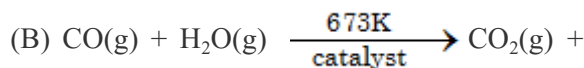
A, B are wheatstone so no change.

PART-2 : CHEMISTRY

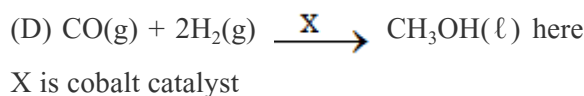
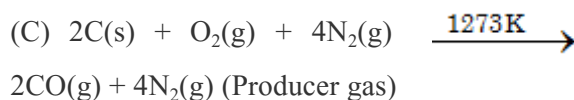
SECTION-I

3. Ans (A,B,C,D)

(A) Water gas is prepared by $C(s) + H_2O$



it is known as water gas shift reaction



4. Ans (A,B,C)

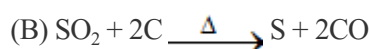
(A) Bigger anion is stabilised by bigger cation through lattice energy effect.

(B) Because of their high reactivity towards air and water on account of their higher electropositive character

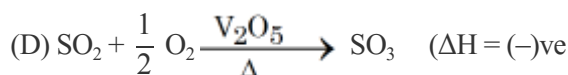
(C) All alkali metals are highly electropositive in nature and, therefore, their hydrides are ionic solids with high melting points.

(D) In highly concentrated solution, unpaired electrons with opposite spins paired up-forming the solution diamagnetic.

5. Ans (A,B,C,D)



(C) P_2O_5 is a better dehydrating agent than H_2SO_4



SECTION-II (i)

1. **Ans (0.04)**

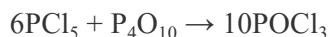
$$\pi = CRT \Rightarrow 16.628 \times 10^3 = C \times 8.314 \times 300$$

$$\therefore C = \frac{20}{3} \text{ mol/m}^3$$

$$= \frac{20}{3 \times 10^6} \times 100 \times 60\% (\text{w/v})$$

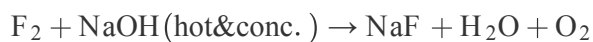
$$= 0.04 \% (\text{w/v})$$

3. **Ans (1.25)**



SECTION-II (ii)

6. **Ans (8.00)**



7. **Ans (5.00)**

1, 2, 5, 6, 7

8. **Ans (5.00)**

2, 3, 4, 5, 6

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,B,C)**

$$|a| = |b| = |c| = \sqrt{3}$$

$$a \wedge b = a \wedge c = b \wedge c = 60^\circ$$

$$\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = x\vec{a} + y\vec{b} + z\vec{c}$$

$$(A) [\vec{a} \vec{b} \vec{c}] = 3x + \frac{y}{2} + \frac{z}{2}$$

$$(B) [\vec{a} \vec{b} \vec{c}] = \frac{3}{2}x + 3y + \frac{3}{2}z$$

$$(C) 0 \Rightarrow \frac{3}{2}x + \frac{3}{2}y + 3z$$

$$\Rightarrow x = y = -z$$

$$(\text{Volume})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 3/2 & 3/2 \\ 3/2 & 3 & 3/2 \\ 3/2 & 3/2 & 3 \end{vmatrix} = \frac{27}{2}$$

$$[\vec{a} \vec{b} \vec{c}] = \frac{3\sqrt{3}}{\sqrt{2}} \Rightarrow x = \frac{\sqrt{3}}{\sqrt{2}}$$

$$x^2 + y^2 + z^2 = 3x^2 = \frac{3 \cdot 3}{2} = \frac{9}{2} < \sqrt{21}$$

2. **Ans (A,C,D)**

at $x = 1$; $f(1^-) = 1$, $f(1^+) = 1$ also $f(1) = 1$.

$$f(x) = \begin{cases} x^3 (\text{sgn}[x] + \{x\}) & ; 0 \leq x < 1 \\ 1 & ; x = 1 \\ x^3 (\text{sgn}[x] + \{x\}) & ; 1 < x < 2 \\ \cos x + |x - 2| & ; 2 \leq x < \pi \end{cases}$$

$$f(x) = \begin{cases} x^4 & ; 0 \leq x < 1 \\ 1 & ; x = 1 \\ x^3(x) & ; 1 < x < 2 \\ \cos x + x - 2 & ; 2 \leq x < \pi \end{cases}$$

at $x = 1$, $f(1^-) = f(1^+) = f(1) = 1 \Rightarrow$ continuous

at $x = 2$ $f(2^-) = 16$, $f(2^+) = \cos 2$.

$\Rightarrow x = 2$ discontinuous $\Rightarrow$ NOT differentiable.

3. **Ans (C)**

$$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

dot product with $\vec{a}$, $\vec{b}$, $\vec{a} \times \vec{b}$ we get

$$4x + y = 2$$

$$x + y = 8$$

$$3z = 4\sqrt{3}$$

$$x = -2, y = 10, z = \frac{4}{\sqrt{3}}$$

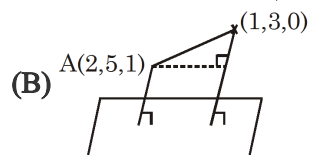
4. **Ans (A,B,D)**

$$(A) \vec{n}_1 \times \vec{n}_2 = \langle -2, 3, -1 \rangle$$

point common to P_1 and P_2 is $(1, 2, 0)$

$\therefore$ distance between skew lines is

$$\frac{|(\vec{a} - \vec{c}) \cdot (\vec{p} \times \vec{q})|}{|\vec{p} \times \vec{q}|} = \frac{1}{5\sqrt{3}}$$

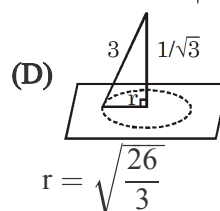


$$\vec{BA} = \hat{i} + 2\hat{j} + \hat{k}$$

angle between line and plane =

$$\sin^{-1} \left(\frac{4}{\sqrt{3}\sqrt{6}} \right) = \theta$$

$$\text{projection is } |\vec{BA}| \cos \theta = \frac{\sqrt{6} \times \sqrt{2}}{\sqrt{18}} = \frac{\sqrt{6}}{3}$$



$$r = \sqrt{\frac{26}{3}}$$

$$\text{area} = r = \frac{26\pi}{3}$$

5. Ans (B,C,D)

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{e^{\tan^{-1}h} + h^{2017} \cdot \sin^2 \frac{1}{h} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \left(\left(\frac{e^{\tan^{-1}h} - 1}{\tan^{-1}h} \cdot \frac{\tan^{-1}h}{h} \right) + \left(h^{2016} \cdot \sin^2 \frac{1}{h} \right) \right)$$

$$= 1 + 0 = 1$$

similarly $f'(1^-) = 1 \Rightarrow f'(1) = 1$

Now, $\ell = \lim_{n \rightarrow \infty} \left(n \sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - 10n \right)$

$$= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - 10 \right)$$

$$= \lim_{n \rightarrow \infty} \frac{f\left(1 + \frac{1}{n}\right) - f(1)}{\frac{1}{n}} + 2 \left(\frac{f\left(1 + \frac{2}{n}\right) - f(1)}{\frac{2}{n}} \right) +$$

$$\dots + 10 \left(\frac{f\left(1 + \frac{10}{n}\right) - f(1)}{\frac{10}{n}} \right)$$

$$= f'(1) + 2f'(1) + \dots + 10f'(1) = 55f'(1) = 55$$

6. Ans (A,B)

$f^{-1}(x) = xe^x = g(x)$ (say) $\Rightarrow f(x) = g^{-1}(x)$

clearly, $g(e) = e^{(e+1)} \Rightarrow g^{-1}\left(e^{(e+1)}\right) = e$

$\Rightarrow f\left(e^{(e+1)}\right) = e \Rightarrow f\left(f\left(e^{(e+1)}\right)\right) = f(e)$

$= g^{-1}(e) = 1$

Now, $ye^y = x$; $y = f(x)$

So, $\ell ny + y = \ell nx \Rightarrow \frac{y'}{y} + y' = \frac{1}{x}$

$\Rightarrow f'(x) = \frac{y}{x(y+1)}$

$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{\ell nx} = \lim_{x \rightarrow \infty} x f'(x) = \lim_{x \rightarrow \infty} \frac{f(x)}{f(x)+1} = 1$

(as $f(x) \rightarrow \infty$ when $x \rightarrow \infty$)

7. Ans (A,D)

$$f(x) = \begin{cases} \frac{1 - \cos\left(1 - \tan\left(\frac{\pi}{4} - x\right)\right)}{x^2}, & x > 0 \\ \frac{\lambda \sin 4x}{x}, & x < 0 \end{cases}$$

$$f(0^+) = \lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \tan\left(\frac{\pi}{4} - x\right)\right)}{x^2}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(1 - \tan\left(\frac{\pi}{4} - x\right)\right)^2}{x^2} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{4 \tan^2 x}{x^2 (1 + \tan x)^2} = 2$$

$$f(0^-) = \lim_{x \rightarrow 0^-} \frac{\lambda \sin 4x}{x} = 2 \Rightarrow \lambda = \frac{1}{2}$$

$\therefore f(0) + 2\lambda = 3.$

8. Ans (B,C)

$y = x^{n-1} \ell n x$

$y_1 = \frac{x^{n-1}}{x} + (n-1)x^{n-2} \ell n x$

$y_1 = x^{n-2} + (n-1) \cdot \frac{y}{x} \quad \dots (1)$

$xy_1 = x^{n-1} + (n-1)y$

$xy_2 + y_1 = (n-1)x^{n-2} + (n-1)y_1 \quad \dots (2)$

$xy_2 + y_1 = (n-1)\left(y_1 - (n-1)\frac{y}{x}\right) + (n-1)y_1$

$xy_2 + y_1 = (n-1)y_1 - (n-1)^2 \frac{y}{x} + (n-1)y_1$

$x^2y_2 + xy_1 = (n-1)xy_1 - (n-1)^2y + (n-1)xy_1$

$x^2y_2 + xy_1 (1 - n + 1 - n + 1) + (n-1)^2y = 0$

$x^2y_2 + xy_1 (3 - 2n) + (n-1)^2y$

$f(n) = 3 - 2n$

$g(n) = n^2 - 2n + 1$

$f(n) - g(n) = 2 - n^2$

$f(3) - g(3) = 2 - 9 = -7$

$f(4) + g(4) = -5 + 9 = 4$

$f(5) \cdot g(5) = (-7)(16) = -112$

SECTION-II (i)

1. Ans (3.00)

$x = -2, x = 3, x = 1$ are three such values

2. Ans (0.25)

$$f(x) = \sqrt{\cos^{-1}(2x) + 2} + \sqrt{1 - \cos^{-1}(2x)}$$

$$1 - \cos^{-1}(2x) \geq 0$$

$$\Rightarrow -1 \leq 2x \leq 1 \quad \cos^{-1}(2x) \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2} \quad 2x \geq \cos(1)$$

$$x \geq \frac{1}{2} \cos 1$$

$$\text{Domain is } \frac{1}{2} \cos 1 \leq x \leq \frac{1}{2}$$

$$\& \text{ Range : } [\sqrt{3}, \sqrt{2} + 1]$$

$$a + b = 5$$

3. Ans (0.80)

The largest integer in the given interval is $\left\lceil \frac{3x}{10} \right\rceil$.

$$\text{If we put } \frac{3x}{10} = y \text{ then } [y] - 3 < \frac{y}{3} \leq [y] - 2$$

solving;

$$y \in \{3\} \cup [4, 5) \Rightarrow x \in \{10\} \cup \left[\frac{40}{3}, \frac{50}{3} \right)$$

So the possible integral values of x are 10, 14, 15, 16

4. Ans (0.33)

$$f(x) + f(1-x) = 1$$

$$f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) +$$

$$\dots f\left(\frac{1}{2}\right) + f\left(1 - \frac{49}{100}\right)$$

$$\dots f\left(1 - \frac{1}{100}\right) = 49 + \frac{1}{2} = \frac{99}{2}$$

$$\int_0^2 \frac{99}{2} \cdot dx = 99$$

SECTION-II (ii)

5. Ans (0.25)

$$f(x) = \cos^{-1} \left(\frac{2 - (x+1)^2}{(x+1)^2} \right)$$

$$= \cos^{-1} \left(\frac{2}{(x+1)^2} - 1 \right)$$

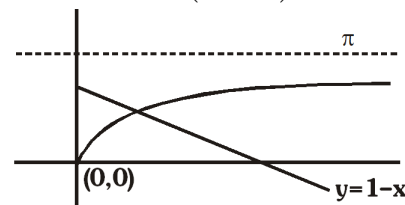
$$\left(\text{Let } \frac{1}{x+1} = \cos \theta \right)$$

$$= \cos^{-1} (2\cos^2 \theta - 1)$$

$$= \cos^{-1} (\cos 2\theta)$$

$$= 2\theta \quad (\text{because } x > 0 \Rightarrow 0 < \cos \theta \leq 1)$$

$$f(x) = 2\cos^{-1} \left(\frac{1}{x+1} \right)$$



Number of solutions are 1

6. Ans (6.00)

$$|\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2$$

$$= 2(\vec{a}^2 + \vec{b}^2 + \vec{c}^2 + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$= 2(9 + 16 + 25 + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{a} = -\vec{a}^2 = -9$$

$$\text{Also } |\vec{b} + \vec{c}|^2 = 9$$

$$\Rightarrow 16 + 25 + 2\vec{b} \cdot \vec{c} = 9$$

$$\Rightarrow \vec{b} \cdot \vec{c} = -16$$

$$\therefore |\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2$$

$$2(50 + 16 - 9) = 114$$

7. Ans (2.00)

$$\lim_{x \rightarrow 0} e^{\left(3 + \frac{f(x)}{x^2}\right) \frac{1}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(2 + \frac{ax^3 + bx^2 + cx + d}{x^2}\right) \frac{1}{x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{ax^3 + (b+2)x^2 + cx + d}{x^3}\right) = 2$$

$$\Rightarrow a = 2, b = -2, c = 0, d = 0$$

8. Ans (1.00)

Given

$$f(x) \cdot f(y)$$

$$+ f\left(\frac{a}{x}\right) \cdot f\left(\frac{a}{y}\right) = 2f(xy) \forall x, y \in \mathbb{R}^+ \dots (1)$$

$$\text{put } x = y = 1 \text{ in (1)}$$

$$\text{then (1)} \Rightarrow f^2(1) - 2f(1) + 1 = 0 \Rightarrow f(1) = 1$$

$$\text{put } y = 1 \text{ in (1)}$$

$$f(x) + f\left(\frac{a}{x}\right) = 2f(x) \Rightarrow f(x) = f\left(\frac{a}{x}\right) \dots (2)$$

$$\text{put } y = \frac{a}{x} \text{ in (1)}$$

$$(1) \Rightarrow f(x) \cdot f\left(\frac{a}{x}\right) + f(x) \cdot f\left(\frac{a}{x}\right) = 2f(a) = 2$$

$$\Rightarrow f(x) f\left(\frac{a}{x}\right) = 1$$

$$\text{using (2), } f^2(x) = 1$$

$$\Rightarrow f(x) = 1 \forall x \in \mathbb{R}^+ (\because f(1) = 1)$$

$$\therefore f(1) - (f(2))^2 + \sqrt{f(3)}$$

$$= 1 - 1 + 1 = 1$$

SECTION-IV

1. Ans (A → P, B → Q, C → S, D → Q)

$$(A) \tan(2\pi |\sin \theta|) = \tan\left(\frac{\pi}{2} - |\cos \theta| 2\pi\right)$$

$$\Rightarrow |\sin \theta| + |\cos \theta| = \frac{n}{2} + \frac{1}{4} \dots (1)$$

$$\Rightarrow \sin \theta \quad 1 \leq |\sin \theta| + |\cos \theta| \leq \sqrt{2}$$

then

$$1 \leq \frac{n}{2} + \frac{1}{4} \leq \sqrt{2} \Rightarrow \frac{3}{2} \leq n \leq \frac{4\sqrt{2}-1}{2} \text{ then}$$

$n = 2$ is only possible value.

$$g(x) = \lim_{n \rightarrow \infty} \left[2 \left(\frac{4}{5}\right)^x\right] = 0$$

$$(B) \tan^2 \alpha + \cot^2 \alpha$$

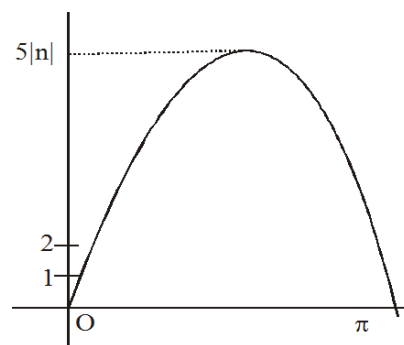
$$= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 8x + 3} - \sqrt{x^2 + 4x + 2}\right) = 2$$

$$\text{given } \tan^2 \alpha + \frac{1}{\tan^2 \alpha} = 2$$

$$\Rightarrow \tan^2 \alpha = 1 \Rightarrow \alpha = \pm \frac{\pi}{4}$$

$$\text{then } \frac{1}{2}(\tan \alpha + \cot \alpha) = 1, -1$$

$$(C) [2 + 5|n| |\sin x|] = 2 + [5|n| |\sin x|]$$



Number of points of non derivability

$$= 2(5|n|-1) + 1 = 10|n|-1 = 19$$

$$\Rightarrow |n| = 2$$

(D) Adding equation (1) and (2)

$$(\sin^{-1} y)^2 = \frac{4n+1}{32} \pi^2$$

$$\Rightarrow 0 \leq \frac{4n+1}{32} \pi^2 \leq + \frac{\pi^2}{4} \Rightarrow \frac{-1}{4} \leq n \leq \frac{7}{4} \dots (3)$$

$$\text{Also } \cos^{-1} x = \frac{4n-1}{32} \pi^2$$

$$\Rightarrow \frac{1}{4} \leq n \leq \frac{8}{\pi} + 6 \dots (4)$$

From equation (3) and (4) $n = 1$

2. Ans (A → Q, B → R, C → Q, D → P)

$$(A) \bar{a} \times (\bar{a} \times \bar{b}) = (\bar{a} \cdot \bar{b}) \bar{a} - (\bar{a} \cdot \bar{a}) \bar{b}$$

$$\bar{a} \times (\bar{a} \times (\bar{a} \times \bar{b})) = -(\bar{a} \times \bar{b})$$

$$\bar{a} \times (\bar{a} \times (\bar{a} \times (\bar{a} \times (\bar{a} \times \bar{b})))) = (\bar{a} \times \bar{b})$$

Similarly if $(4n + 1)$ $\bar{a}$'s are there then product is $(\bar{a} \times \bar{b})$.

Hence $m = 1$.

(Q) Clearly the value is least when P is centroid of tetrahedron.

(R) Ceva's theorem.

$$(S) \text{adj}(\text{adj} A) = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$\Rightarrow |\text{adj}(\text{adj} A)|^2 = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$$

$$= [\vec{a}, \vec{b}, \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{vmatrix} = \frac{1}{2}$$

$$\Rightarrow |A|^8 = \frac{1}{2}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

15-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C,D	D	A,B,D	A,C	A,B,C	B,C	C,D	C
SECTION-I (iii)	Q.	13	14	15	16	17	18		
	A.	B	B	A,D	A,B,C	A,C	B		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C	A,B,C	A	A,C,D	A,B,D	A,B,C	A,C,D	B,C,D
SECTION-I (iii)	Q.	13	14	15	16	17	18		
	A.	B	A	A,B,C	A,B	A,D	A,C,D		

PART-3 : MATHEMATICS

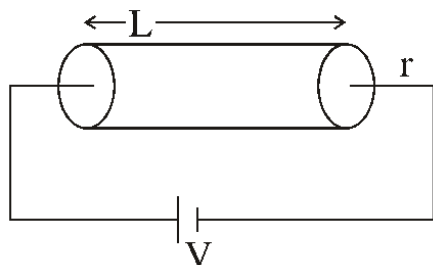
SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B	A,D	A,B,C	A,B,D	A,C	A,B,C	A,B,D	B,C,D
SECTION-I (iii)	Q.	13	14	15	16	17	18		
	A.	A,B	C	A	C	A,C,D	C		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (D)



$$i = \frac{V}{R} = \frac{V A}{\rho L} = neAV_d,$$

$$V_d = \frac{V}{ne\rho L}$$

V_d is independent of area of cross section.

2. Ans (B)

$$5 \text{ V.S.D} = 4 \text{ MSD}$$

$$1 \text{ VSD} = \frac{4}{5} \text{ MSD}$$

$$\text{L.C.} = 1 \text{ MSD} - 1 \text{ VSD} = \frac{1}{5} \text{ MSD} = 1 \text{ mm}$$

$$\text{For screw gauge LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$$

3. Ans (B)

% error

$$= 100 \left(\frac{\Delta \ell}{\ell} + \frac{4 \Delta r}{r} \right) = \left(\frac{0.1}{10} + 4 \left(\frac{0.01}{2} \right) \right) \times 100 = 3$$

4. **Ans (B)**

At point P

$$a < r < b$$

$$E_P = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$V_P = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} - \frac{Q}{b} \right]$$

$$V_A - V_B = \left(\frac{kQ}{a} - \frac{kQ}{b} \right) - 0$$

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

SECTION-I (ii)

5. **Ans (B,C,D)**

$$\Delta d = \Delta \ell = \frac{0.5}{100} \text{ mm}$$

$$y = \frac{4MLg}{\pi \ell d^2}$$

$$\left(\frac{\Delta y}{y} \right)_{\max} = \frac{\Delta \ell}{\ell} + 2 \frac{\Delta d}{d}$$

error due to ℓ measurement

$$\frac{\Delta \ell}{\ell} = \frac{0.5/100 \text{ mm}}{0.25 \text{ mm}}$$

error due to d measurement

$$2 \frac{\Delta d}{d} = \frac{2 \times \frac{0.5}{100}}{0.5 \text{ mm}} = \frac{0.5/100}{0.25}$$

So error in y due to ℓ measurement = error in y due to d measurement

6. **Ans (D)**

$$\text{If } E_2 > E_1; E_1 < \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} < E_2$$

7. **Ans (A,B,D)**

$$0 \leq x \leq a; V_x$$

$$= \left[- \int_0^x E_x dx \right] + V_{(0)} = 0 \quad (\text{as } E_x = 0)$$

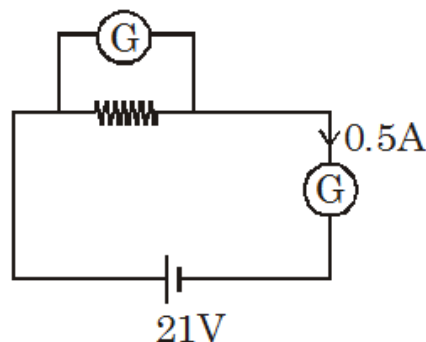
$$x \geq a; V_x = - \int_a^x E_x dx + V_{(a)}$$

$$= \left[- \int_a^x \frac{\sigma}{\epsilon_0} dx \right] + V_{(a)} = - \frac{\sigma}{\epsilon_0} (x - a)$$

$$x \leq 0; V_x = - \int_a^x E_x dx + V_{(0)}$$

$$= - \left(- \frac{\sigma}{\epsilon_0} \cdot x \right) + V_{(0)} = \frac{\sigma}{\epsilon_0} \cdot x$$

8. **Ans (A,C)**



$$\frac{2100}{42} = R + G$$

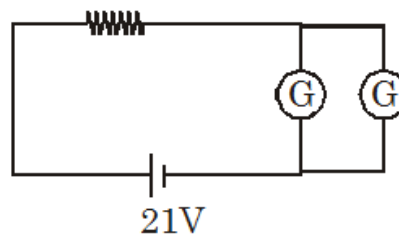
$$R_{eq} = \frac{21}{0.5} = 42\Omega = \frac{RG}{R+G} + G$$

$$\Rightarrow 8 = R - \frac{RG}{R+G} = \left(\frac{R^2}{R+G} \right)$$

$$8 \times 50 = R^2$$

$$R = 20\Omega \Rightarrow G = 30\Omega$$

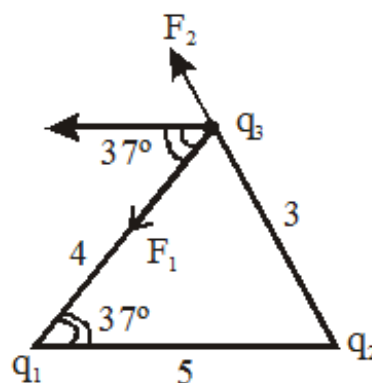
$$i_2 = 0.5 \times \frac{R_{eq}}{G} = 0.5 \times \frac{20 \times 30}{20+30} = 0.2 \text{ A}$$



$$R_{eq} = (20 + 15)\Omega = 35\Omega.$$

$$i = \frac{21}{35} = 0.6\text{A} \Rightarrow i_{\text{each}} = 0.3 \text{ A}$$

9. **Ans (A,B,C)**



As per the given condition ' q_1 ' must be negative & ' q_2 ' must be positive

$$F_1 \cos 53 = F_2 \cos 37$$

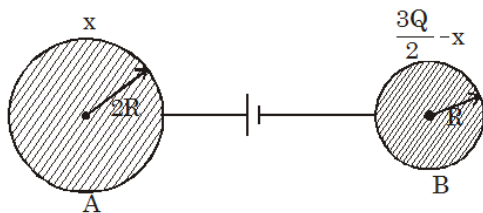
solving this we get

$$q_2 = \frac{27}{32} \mu\text{C}$$

11. Ans (C,D)

Total charge will be conserved as total charge supplied by battery is always zero.

let final charge is as



$$V_A = V_B + v$$

$$\frac{kx}{2R} = \frac{k\left(\frac{3Q}{2} - x\right)}{R} + v$$

$$\frac{3kx}{2R} = v + \frac{3kQ}{2R}$$

$$x = \frac{2Rv}{3k} + Q > \frac{Q}{2}$$

12. Ans (C)

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

% error

$$= \frac{\Delta R_1}{R_1} \times 100 + \frac{\Delta R_2}{R_2} \times 100 + \frac{\Delta(R_1 + R_2)}{(R_1 + R_2)} \times 100$$

$$= 1 + 2 + \frac{.03 + 0.12}{9} \times 100$$

$$= 3 + 1.67 = 4.67\%$$

SECTION-I (iii)

13. Ans (B)

$$\frac{\sigma}{2\epsilon_0} ea = \frac{1}{2} mv^2$$

$$v = \sqrt{\frac{e\sigma a}{m\epsilon_0}}$$

14. Ans (B)

$$\text{acceleration} = \frac{e\sigma}{2\epsilon_0 m}$$

$$v_0 = \frac{e\sigma}{2\epsilon_0 m} t$$

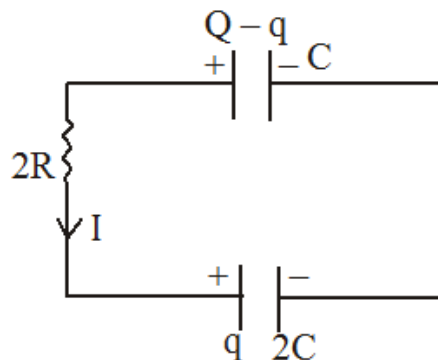
$$t = \frac{2v_0 m\epsilon_0}{e\sigma}$$

$$T = 4t$$

$$T = \frac{8v_0 m\epsilon_0}{e\sigma}$$

15. Ans (A,D)

at 't' time



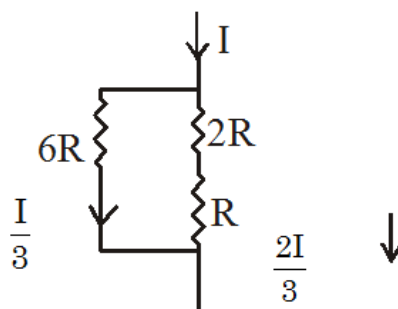
$$\frac{Q - q}{C} - 2IR - \frac{q}{2C} = 0$$

$$\frac{2Q - 3q}{4C} = \frac{dq}{dt} R$$

$$\int_0^q \frac{dq}{2Q - 3q} = \int_0^t \frac{dt}{4RC}$$

$$\ln\left(\frac{2Q - 3q}{2Q}\right) = -\frac{3t}{4RC}$$

$$q = \frac{2Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$$



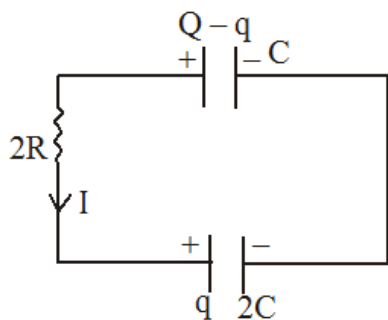
$$\text{total power} = 2I^2 R = P$$

$$\text{Power of resistance} = \frac{4I^2}{9} R = \frac{2}{9} P$$

$$\text{so, heat product in } R = \frac{2}{9} H = \frac{2}{9} \left(\frac{Q^2}{3C}\right) = \frac{2Q^2}{27C}$$

16. Ans (A,B,C)

at 't' time



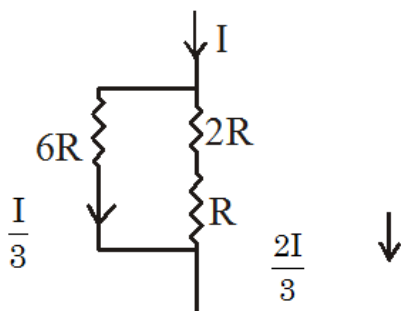
$$\frac{Q-q}{C} - 2IR - \frac{q}{2C} = 0$$

$$\frac{2Q-3q}{4C} = \frac{dq}{dt} R$$

$$\int_0^q \frac{dq}{2Q-3q} = \int_0^t \frac{dt}{4RC}$$

$$\ln\left(\frac{2Q-3q}{2Q}\right) = -\frac{3t}{4RC}$$

$$q = \frac{2Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$$



$$\text{total power} = 2I^2R = P$$

$$\text{Power of resistance} = \frac{4I^2}{9} R = \frac{2}{9} P$$

$$\text{so, heat product in } R = \frac{2}{9} H = \frac{2}{9} \left(\frac{Q^2}{3C}\right) = \frac{2Q^2}{27C}$$

17. Ans (A,C)

$$g = \frac{GM}{R^2} = \frac{6.6742 \times 10^{-11} \times 6 \times 10^{24}}{6.4^2 \times 10^{12}} = 9.77666$$

$$\frac{dg}{g} = \frac{dG}{G} + \frac{dM}{M} + \frac{2dR}{R}$$

$$= \frac{0.001}{6.6742} + 0.0001 + 2 \times 0.0002 = 0.00065$$

$$\therefore g = (9.77767 \pm 0.0007) \text{ m/s}^2$$

5 significant figure

18. Ans (B)

$$v = \sqrt{\frac{GM}{R}}$$

$$\frac{dv}{v} = \frac{1}{2} \frac{dM}{M} + \frac{1}{2} \frac{dR}{R} + \frac{1}{2} \frac{dG}{G}$$

$$= \frac{1}{2} \left[0.01 + 0.02 + \frac{0.001}{6.6742} \times 100 \right] \%$$

$$= 0.0225\%$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)

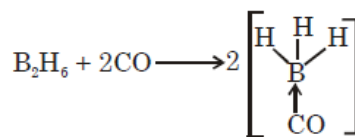
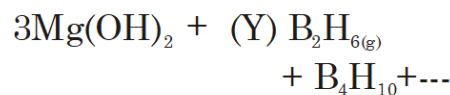
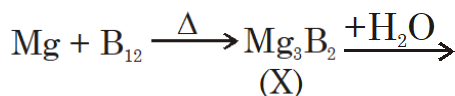
$$P_t = P_A^0 x_A + P_B^0 x_B = \frac{1}{4} = P_A^0 + \frac{3}{4} \times 3P_A^0$$

$$= \frac{10}{4} P_A^0 = \frac{5}{2} P_A^0$$

$$P_t \cdot Y_A = P_A^0 x_A$$

$$\Rightarrow y_A = \frac{P_A^0 \times \frac{1}{4}}{\frac{5}{2} P_A^0} = \frac{1}{10}$$

2. Ans (A)



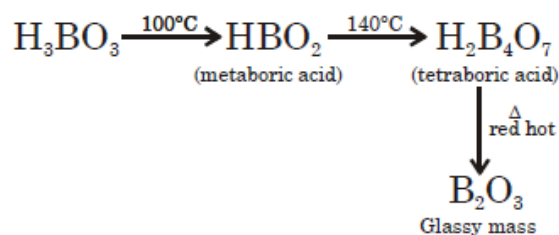
SECTION-I (ii)

8. Ans (A,C,D)

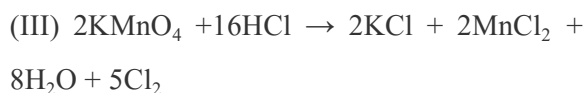
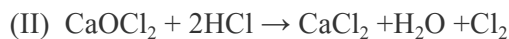
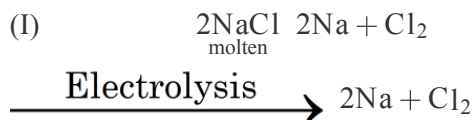
Acidic nature : HF < HCl < HBr < HI

Boiling point : HCl < HBr < HI << HF

9. Ans (A,B,D)

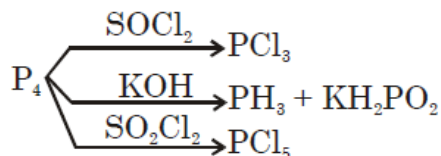
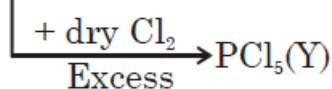
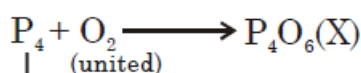


10. Ans (A,B,C)

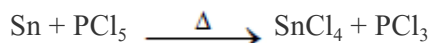


SECTION-I (iii)

15. Ans (A,B,C)



16. Ans (A,B)

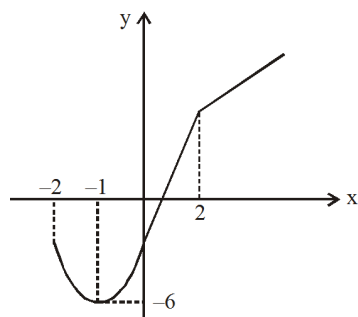


PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$f(x) = \begin{cases} 2x^2 + 4x - 4 & x \in [-2, 2] \\ 4x + 4 & x \in (2, \infty) \end{cases}$$



$\therefore$ Range : $[-6, \infty)$

2. Ans (C)

$$\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\alpha = \frac{\vec{r} \cdot (2\hat{i} + \hat{k})}{\sqrt{5}} = \frac{2a + b}{\sqrt{5}}$$

$$\beta = \frac{-2a + b}{\sqrt{5}}, \quad r = c$$

$$\therefore |\alpha| = |\beta| = |\gamma|$$

$$\Rightarrow$$

$$\underbrace{|2a + b|}_A = \underbrace{|-2a + b|}_B = |c\sqrt{5}|$$

$$\left. \begin{matrix} a = \pm \frac{\sqrt{5}}{3} \\ b = 0 \\ c = \pm \frac{2}{3} \end{matrix} \right\} \rightarrow 4$$

$$\left. \begin{matrix} c = \pm \sqrt{\frac{1}{6}} \\ b = \pm \sqrt{\frac{5}{6}} \\ a = 0 \end{matrix} \right\} \rightarrow 4$$

8 possible triplets.

3. Ans (B)

Applying c and d

$$\frac{\pi/2 + 2 \times \pi/2}{\sin^{-1}x^{2018} - \cos^{-1}x^{2018} + 2\{\cos^{-1}x^{2017} - \sin^{-1}x^{2017}\}} = \frac{-3}{-7}$$

$$\frac{3\pi/2}{4\cos^{-1}x^{2017} - 2\cos^{-1}x^{2018} - \frac{\pi}{2}} = \frac{3}{7}$$

$$\Rightarrow \frac{7\pi}{2} = 4\cos^{-1}x^{2017} - 2\cos^{-1}x^{2018} - \frac{\pi}{2}$$

$$2\cos^{-1}x^{2017} = 2\pi + \cos^{-1}x^{2018}$$

$$x = 1 \text{ \& } -1$$

$x = -1$ satisfies original equation

$$x = -1$$

$$\text{so, } \frac{\sin^{-1}(\sin \alpha) + \tan^{-1}(\tan \alpha)}{\cos^{-1}(\cos \alpha) + 2 \cos^{-1}(\cos \alpha)} = 2$$

4. Ans (B)

Domain $|x| < 1$

Let $|x| = \sin \theta$ in 1st element

$$T_1 = \tan^{-1} \tan \theta = \sin^{-1} |x|$$

$$|x| = \tan \theta \text{ in 2nd element}$$

$$T_2 = \cos^{-1} \cos \theta = \tan^{-1} |x|$$

Now $f(x) = \sin^{-1} |x| + \tan^{-1} |x|$ (even and increasing function in $[0, 1)$)

$$\text{range of } f(x) \text{ is } \left[0, \frac{3\pi}{4}\right) \Rightarrow \text{three integers } 0, 1, 2.$$

SECTION-I (ii)

5. Ans (B)

$$\text{Domain of } f(x) = \mathbb{R} - \left[-\frac{d}{c} \right]$$

$$\text{Range of } f(x) = \mathbb{R} - \left[\frac{a}{c} \right]$$

since $f(x)$ is inverse of itself

Domain = Range

$$\Rightarrow d = -a$$

$$f(x) = \frac{ax + b}{cx - a}$$

$$f(19) = 19, f(97) = 97$$

$$\Rightarrow \frac{a}{c} = 58$$

6. Ans (A,D)

$$\tan^{-1}x + \tan^{-1}\frac{1}{y} = \tan^{-1}3$$

$$\frac{x + \frac{1}{y}}{1 - \frac{x}{y}} = 3$$

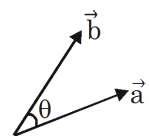
$$\Rightarrow xy + 1 = 3y - 3x$$

$$\Rightarrow x = \frac{3y-1}{y+3} = 3 - \frac{10}{y+3}$$

$$y+3 = 5, 10$$

$$y = 2, 7 \Rightarrow x = 1, 2$$

7. Ans (A,B,C)



$$\cos \theta = \frac{11}{14}$$

$$\vec{a} = \hat{i} + x\hat{j} + 3\hat{k}$$

$$\vec{b} = 4\hat{i} + (4x-2)\hat{j} + 2\hat{k}$$

$$2|\vec{a}| = |\vec{b}|$$

$$4(1+x^2+9) = (16+(4x-2)^2+4)$$

$$3x(x-2) + 2(x-2) = 0$$

$$x = 2, x = -\frac{2}{3}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{11}{14}$$

$$\frac{4+x(4x-2)+6}{2(10+x^2)} = \frac{11}{14}$$

$$17x^2 - 14x - 40 = 0$$

$$x = 2$$

8. Ans (A,B,D)

$$(\vec{a} \times \vec{\beta}) \times (\vec{\gamma} \times \vec{a}) + \vec{a} + \vec{\beta} = \vec{\beta} \times \vec{\gamma}$$

$$\text{Let } \vec{k} = \vec{a} \times \vec{\beta}$$

$$\Rightarrow \vec{k} \times (\vec{\gamma} \times \vec{a}) + (\vec{a} + \vec{\beta}) = \vec{\beta} \times \vec{\gamma}$$

$$\Rightarrow \vec{\gamma} (\vec{k} \cdot \vec{a}) - \vec{a} (\vec{k} \cdot \vec{\gamma}) + \vec{a} + \vec{\beta} = \vec{\beta} \times \vec{\gamma}$$

$$\Rightarrow 0 - \vec{a} (\vec{k} \cdot \vec{\gamma}) + \vec{a} + \vec{\beta} = \vec{\beta} \times \vec{\gamma}$$

$$\Rightarrow -\vec{a} [\alpha\beta\gamma] + \vec{a} + \vec{\beta} = \vec{\beta} \times \vec{\gamma}$$

$$\Rightarrow \vec{a} (-[\alpha\beta\gamma] + 1) + \vec{\beta} = \vec{\beta} \times \vec{\gamma} \quad \dots(i)$$

Taking dot product with in equation (i)

$$\Rightarrow \vec{\gamma} \cdot \vec{a} (\lambda) + \vec{\gamma} \cdot \vec{\beta} = 0$$

$$\Rightarrow \vec{\gamma} \cdot (\vec{a}k - \vec{\beta}) = 0 \Rightarrow \vec{a}k = \vec{\beta}$$

$\vec{a}$ is parallel to $\vec{\beta}$

Now cross product with $\vec{\beta}$ in equation (i)

$$\Rightarrow \vec{\beta} \times \vec{a} (\mu) + \vec{\beta} \times \vec{\beta} = \vec{\beta} (\vec{\beta} \times \vec{\gamma})$$

$$\Rightarrow 0 + 0 = \vec{\beta} (\vec{\beta} \cdot \vec{\gamma}) - \vec{\gamma} (\vec{\beta} \cdot \vec{\beta})$$

$$\vec{\gamma} = \lambda\vec{\beta}$$

9. Ans (A,C)

$$|xy| - r|x| - r|y| + r^2 \leq 0$$

$$\Rightarrow (|x| - r) (|y| - r) = 0$$

$$P(r) = (2r+1)^2$$

$$\sum_{r=1}^{r=n} P(r) = \frac{n}{3} (4n^2 + 12n + 11)$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n}{3} (4n^2 + 12n + 11) - \lambda n^3}{an^2 + bn + c} = \frac{1}{2}$$

$$\Rightarrow \lambda = \frac{4}{3} \text{ and } a = 8$$

10. Ans (A,B,C)

(A) $g(x) = x^7 - f(x)$

$g(0) = -f(0) < 0$

$g(1) = 1 - f(1) > 0$

$\Rightarrow g(x) = 0$ for some $x \in (0, 1)$

(B) $x^2 - \int_0^x f(t) dt = g(x)$

$g(0) = 0, g(1) = 1 - \int_0^1 f(t) dt > 0$

So, $g(x)$ can be positive in $(0, 1)$.

(C) $g(x) = \int_0^{\pi/2-x} f(t) \sin 2t dt - x$

$g(0) = \int_0^{\pi/2} f(t) \sin 2t dt > 0$

$g(1) = \int_0^{\pi/2-1} f(t) \sin 2t dt - 1 < \left(\frac{\pi}{2} - 1\right) 1 - 1 < 0$

(D) $g(x) = \sqrt{x} - \int_0^{x^2} \sqrt{f(t)} dt$

$g(0) = 0, g(1) = 1 - \int_0^1 \sqrt{f(t)} dt > 0$

So, $g(x) = 0$, not confirmed in $(0, 1)$

11. Ans (A,B,D)

Domain of $f(x) = [-1, 4]$

Range of $f(x) = [-1, 2]$

Range of $f(x) + 1 = [0, 3]$

for domain of $f(|x| - 1)$

$-1 \leq |x| - 1 \leq 4$

$\Rightarrow 0 \leq |x| \leq 5$

$\Rightarrow -5 \leq x \leq 5$

For domain of $f(|x|)$

$\Rightarrow -1 \leq |x| \leq 4$

$\Rightarrow |x| \leq 4$

$\Rightarrow -4 \leq x \leq 4$

12. Ans (B,C,D)

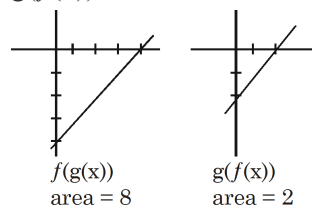
$$f(x) = \begin{cases} -4-2x & x \in [-1, 0) \\ -2x & x \in [0, 1) \\ 4-2x & x \in [1, 2) \\ 8-2x & x \in [2, 3) \end{cases}$$

$$q(x) = \begin{cases} -2-\frac{x}{2} & x \in [-2, 0) \\ -\frac{x}{2} & x \in [0, 2) \\ 2-\frac{x}{2} & x \in [2, 4) \\ 4-\frac{x}{2} & x \in [4, 6) \end{cases}$$

$f(x)$ and $g(x)$ both are bijective

$f(g(x)) = x - 4 \quad \forall x/I$

$g(f(x)) = x - 2 \quad \forall x/I$



SECTION-I (iii)

13. Ans (A,B)

Equating of line L_1 and L_2 can be obtained in symmetric form

$L_1 : \frac{x}{0} = \frac{y-b}{-b} = \frac{z}{c}$

$L_2 : \frac{x-a}{a} = \frac{y}{0} = \frac{z}{c}$

Equation of plane $p : \begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix} = 0$

$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$

Lines L_1 and L_2 in vector form

$L_1 : \vec{r} = b\hat{j} + \lambda(-b\hat{j} + c\hat{k})$

$L_2 : \vec{r} = a\hat{j} + \mu(a\hat{i} + c\hat{k})$

Shortest Distance = $\frac{1}{4} = \frac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}}$

$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 64$

Image of $A(1, 0, 0)$ in the plane

$x - y - z + 1 = 0$ is $\left(\frac{-1}{3}, \frac{4}{3}, \frac{4}{3}\right)$ its

distance from M is 3.

14. **Ans (C)**

Equating of line L_1 and L_2 can be obtained in symmetric form

$$L_1: \frac{x}{0} = \frac{y-b}{-b} = \frac{z}{c}$$

$$L_2: \frac{x-a}{a} = \frac{y}{0} = \frac{z}{c}$$

$$\text{Equation of plane } p: \begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix} = 0$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$$

Lines L_1 and L_2 in vector form

$$L_1: \vec{r} = b\hat{j} + \lambda(-b\hat{j} + c\hat{k})$$

$$L_2: \vec{r} = a\hat{i} + \mu(a\hat{i} + c\hat{k})$$

$$\text{Shortest Distance} = \frac{1}{4} = \frac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}}$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 64$$

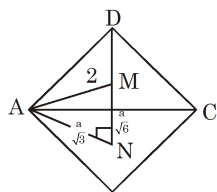
Image of $A(1, 0, 0)$ in the plane $x - y - z + 1 = 0$

is $\left(-\frac{1}{3}, \frac{4}{3}, \frac{4}{3}\right)$ its distance from M is 3

15. **Ans (A)**

$AB = BC = CD = AD = a$

$$a = 2\sqrt{2}$$



$MA = MB = MC = 2$ (using symmetry)

$$MB = 2 \Rightarrow 2b^2 = 4 \text{ or } b = \sqrt{2} (\because b > 0) \therefore$$

$$B(\sqrt{2}, \sqrt{2}, 0)$$

$$MA^2 + MB^2 = AB^2 \Rightarrow \vec{MA} \perp \vec{MB}$$

$$\text{Similarly } \vec{MA} \perp \vec{MC}, \vec{MB} \perp \vec{MC}$$

$$\Rightarrow \vec{MA} \cdot \vec{MC} \Rightarrow \sqrt{2}x + \sqrt{2}y = 0 \text{ and } -2z = 0$$

$$\Rightarrow y = -x \text{ and } z = 0 \therefore C(x, -x, 0). \text{ Also } MC^2 = 2x^2 = MA^2 = 4$$

$$\Rightarrow x = \sqrt{2} (x > 0) \therefore C(\sqrt{2}, -\sqrt{2}, 0)$$

$$\text{Now, coordinates of N are } \left(\frac{2\sqrt{2}}{3}, 0, \frac{-2}{3}\right) \text{ and}$$

hence D is $\left(-\frac{2\sqrt{2}}{3}, 0, \frac{2}{3}\right)$. DN is normal to the plane ABC. N is mid point of DD' so $D' = (2\sqrt{2}, 0, -2)$

16. **Ans (C)**

Since MA, MB, MC are orthogonal triads

$$\therefore \text{volume of tetrahedron is } \frac{1}{6} |MA| \cdot |MB| \cdot |MC| = \frac{4}{3}$$

17. **Ans (A,C,D)**

$$2f(x) \cdot f'(x) - 2f'(x) - \frac{2}{2\sqrt{x}} = 0$$

$$2(f(x) - 1)f'(x) = \frac{1}{2\sqrt{x}}$$

integrating both the sides

$$(f(x) - 1)^2 - \sqrt{x} + C$$

$$\therefore f(0) = 2c = 1$$

$$\Rightarrow (f(x) - 1)^2 = \sqrt{x} + 1$$

$$\Rightarrow f(x) = 1 + \sqrt{\sqrt{x} + 1}$$

$$\text{Now, } \lim_{x \rightarrow 0^+} \frac{f(x) - 2}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{\sqrt{\sqrt{x} + 1} - 1}{\sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x} + 1 - 1}{\sqrt{x}(\sqrt{\sqrt{x} + 1} + 1)} = \frac{1}{2}$$

18. **Ans (C)**

$$\therefore g'(y) = \frac{1}{f'(x)} \Rightarrow g'(4) = \frac{1}{f'(64)}$$

$$(\because f(64) = 4)$$

$$= \frac{1}{\frac{1}{96}} = 96$$